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MILITARY SPECIFICATION

ELECTRICAL DISCHARGE MACHINE, CNC, WIRE

1. SCOPE

1.1 Scope. This specification covers wire cutting computer numerical controlled (CNC), electrical discharge machines with equipment and accessories specified herein.

1.2 Classification. The wire machine shall be of the following sizes. The size to be furnished shall be as specified (see 3.5 and 6.2).

Size	Workpiece dimensions (inches)
4	16 by 16 by 4
5	16 by 22 by 4
6	18 by 27 by 6
7	20 by 16 by 8
8	23 by 19 by 6
9	23 by 35 by 12
10	25 by 29 by 8
11	31 by 22 by 9
12	34 by 24 by 9
13	51 by 47 by 9

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, 2163 Airways Boulevard, Memphis, Tennessee 38114-5051, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interferences.

MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

(Unless otherwise indicated, copies of Federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, (ATTN: Mr. Stephen Lowell), Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

U. S. DEPARTMENT OF LABOR, OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

29 CFR 1910 - Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402-0001.)

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

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2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN GEAR MANUFACTURERS' ASSOCIATION (AGMA)

AGMA 390.03a - Gear Classification, Materials and Measuring Methods for Bevel, Hypoid, Fine Pitch Wormgearing and Racks Only as Unassembled Gears. (DOD adopted)

AGMA 2000-A88 - Gear Classification and Inspection Handbook. Tolerances and Measuring Methods for Unassembled Spur and Helical Gears (DOD adopted)

AGMA 2001-B88 - Fundamental Rating Factors and Calculation Methods for Involute Spur and Helical Gear Teeth.

(Application for copies should be addressed to the American Gear Manufacturers' Association, Standards Department, 1500 King Street, Suite 201, Alexandria, VA 22314-2717.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B5.1 - T-Slots, Their Bolts, Nuts, and Tongues. (DOD adopted)

ANSI/ISO 8860/1 - Dimensional, Physical, and Magnetic Characteristics.

ANSI/ISO 8860/2 - Track Format.

ANSI/ISO 9293 - Volume and File Structure of Flexible Disk Cartridges for Information Interchange.

ANSI/ISO 54 - Cylindrical Gears for General Engineering and for Heavy Engineering, Modules and Diametral Pitches of.

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018-3363.)

ELECTRONIC INDUSTRIES ASSOCIATION (EIA)

RS-227 - One-Inch Perforated Tape.

RS-232 - Data Terminal Equipment and Data Communication Equipment, Interface Between, Employing Serial Binary Data Interchange. (DoD adopted)

RS-244 - Character Control for Numerical Machine Tool Control Perforated Tape. (DOD adopted)

RS-267 - Axis and Motion Nomenclature for Numerically Controlled Machines. (DOD adopted)

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RS-274 - Interchangeable Variable Block Data Format for Positioning, Contouring, and Contouring/Positioning Numerically Controlled Machines. (DOD adopted)

RS-281 - Electrical and Construction Standards for Numerical Machine Control. (DOD adopted)

RS-358 - Subset of American National Standard Code for Information Interchange for Numerical Machine Control Perforated Tape. (DOD adopted)

RS-449 - General Purpose 37-Position and 9-Position Interface for Data Terminal Equipment and Data Circuit-Terminating Equipment Employing Serial Binary Data Interchange.

(Application for copies should be addressed to the Electronic Industries Association, 2001 Eye Street, N.W., Washington, DC 20006-1899.)

MANUFACTURERS' ASSOCIATION (NEMA)

NEMA MG 1 - Motors and Generators. (DOD adopted)

(Application for copies should be addressed to the National Electrical Manufacturers' Association, 2101 L Street, N.W., Washington, DC 20037-1580.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 79 - Electrical Standard for Industrial Machinery. (DOD adopted)

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269-9101.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), one complete wire machine shall be subjected to first article inspection (see 6.4) in accordance with 4.4.

3.2 Design. The wire machine shall be new (not a prototype) and one of the manufacturer's current production models capable of operations in accordance with the requirements herein. The machine shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be readily accessible for adjustment, replacement, or repair.

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3.2.1 Measuring and indicating device calibrations. Measuring and indicating devices shall be graduated in inch-pound (see 6.5.1), metric (see 6.5.2), or dual (inch-pound and metric) units as specified by the procuring activity (see 6.2). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system. All axes positions and movements shall be measured and monitored electronically in increments no larger than 0.0001 in. or 0.001 mm.

3.2.2 Reclaimed materials. The machine may contain reclaimed materials to the maximum extent possible provided such materials will not jeopardize the intended use, performance, or design life of the machine. Reclaimed materials shall have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used, rebuilt, or remanufactured end products are allowed under this specification.

3.2.3 Energy efficiency. The wire machine and its components that directly consume energy in normal operation shall be designed and constructed for energy efficiency in accordance with the developments available within the industry.

3.2.4 Controls. All mechanical, hydraulic, and pneumatic operating controls shall be located convenient to the operator's work station(s).

3.2.5 Safety and health requirements. All parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall conform to all requirements of OSHA 29 CFR 1910. Additional safety and health requirements shall be as specified (see 6.2). Covers, guards, or other safety devices normally furnished as standard on the manufacturer's commercial wire machine supplied to the commercial market shall be provided for the point-of-operation and all other parts of the machine that present safety hazards. If point-of-operation guarding is influenced by workpiece size and configuration, the procuring activity shall provide part design data to the supplier of the machine (see 6.2). The supplier of the machine shall provide the special guarding which shall meet OSHA 29 CFR 1910.

3.2.6 Mercury restriction. The wire machine shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the wire machine.

3.2.8 Environmental protection. During the manufacture, operation, service, transportation, or storage of the machine and its components, materials hazardous to the ecological system as prohibited by applicable Federal, state, or local statutes in effect on the date of the contract shall not be used or emitted.

3.2.9 Lubrication. Means shall be provided to ensure lubrication for all moving parts. Recirculating lubrication systems shall include a cleanable or replaceable filter. Means for determining oil level in each reservoir shall be by an oil level sight gauge, a low oil level warning light, or similar device and shall be in clear view of the operator's normal work station. Each lubricant reservoir shall have at least a 24-hour capacity and means for determining fluid level. All oil holes, grease fittings, and filler caps shall be readily accessible.

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3.2.10 Interchangeability. To provide for replacement of worn parts, all parts shall be manufactured to definite dimensions and tolerances that will permit installation of replacement parts without modification of the part or machine. Replacement parts shall be domestically available.

3.3 Construction. The machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall withstand all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion or failure.

3.3.1 Castings and forgings. All castings and forgings shall be free of defects, scale, and mismatching. No processes such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Fastening devices. All fasteners shall be installed to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be permanently installed.

3.3.3 Surfaces. All surfaces shall be clean and free of harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.4 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. Any excess material used for such operations shall be thoroughly removed from the part(s) upon completion of the operations.

3.3.5 Painting. Unless otherwise specified (see 6.2), the paint color and application shall be in accordance with the manufacturer's commercial practice.

3.3.6 Threads. All threaded parts used on the wire machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Drive train gears. All gears of spindle and axis drive trains shall conform to or exceed all provisions of AGMA 390.03a, AGMA 2000-A88, and AGMA 2001-B88 for the inch-pound system of measurement or ANSI/ISO 54 for the metric system of measurement. The gears shall be manufactured of a suitable metallic material. The gears shall be of proper width and size to transmit full-rated torque and horsepower throughout the speed ranges without failure for the expected service life of the machine. Gears in the drive train shall be hardened and ground steel. Working surface hardness shall be not less than Rockwell C-48.

3.3.8 Electromagnetic interference control. When specified (see 6.2), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystem(s) class and the emission and susceptibility requirements shall be as specified.

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3.4 Components. The wire machine shall consist of not less than a base, work table, wire guide column, wire guides, ways, safety stops, drives, dielectric system, power supply and numerical control system.

3.4.1 Base. The base shall be a box-type ribbed casting with a tensile strength of not less than 34,000 psi, stress relieved and constructed to support any load within the rated capacity of the machine.

3.4.2 Work table. A work table shall be provided on the base that shall be capable of supporting the loads within the rated capacity of the machine. It shall be designed to allow the wire and lower wire guide support to pass through the range of their travel without interference. The table surface shall be provided with threaded holes or T-slots to be used in clamping work pieces. The T-slots shall conform to ANSI B5.1. The table surface shall be machined flat and parallel to the X and Y axes in order to maintain the positioning accuracy required. Provisions shall be made around the table to collect and return dielectric fluid to reservoir/filter.

3.4.3 Wire guide column. The wire support frame shall be designed and constructed to provide adequate support for the wire, wire tensioning and feeding mechanism, wire guides and the U and V axis mechanisms. The wire supply shall be capable of supporting a standard spool of wire. Provisions shall be made to hold used wire. Provisions shall be made to raise and lower the upper wire guide according to the workpiece thickness. The U and V axes shall be located so as to adjust the position of the upper wire guide so that constant tapers may be cut about the periphery of the work piece. Wire tensioning shall be provided from 0 to 1600 grams.

3.4.3.1 Wire guides. Wire guides shall be provided in order to maintain the wire tension and position accuracy. The guides shall be designed to allow the wire to cut at a maximum taper of 20° throughout 360° of X and Y axis.

3.4.4 Ways. The X and Y axes motion shall be provided by table and/or wire guide column motion. Both units shall be machined to accept way bearings of a precise fit to their respective ways. Way bearings shall be replaceable and made of a material with low friction characteristics that will not cause wear to the ways, sticking, overshooting, or put an excessive load on the feed motor and drive components. Antifriction, recirculating roller type way bearings are acceptable. Adjustable gibs, guides, or bearings shall be provided to maintain and adjust alignment of the table to the ways and prevent causing excessive loading of the feed drive motor. Way covers shall be provided.

3.4.5 Safety stops. All machine slide motions shall be provided with a means of overtravel protection to prevent damage to the machine's power feed and rapid traverse mechanisms in the event the machine is inadvertently fed or rapidly traversed into an overtravel limit switch. The machine shall have programmable X and Y travel limits and/or axis inhibit functions to be turned on or off at the control to prevent damage to the machine tool as a result of workpiece or fixturing interference.

3.4.6 Drives. All axis slide motions shall be driven by independent, infinitely variable, direct current (DC) servo type motors, coupled to a precision, anti-friction, recirculating ball screw and nut assembly.

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3.4.7 Dielectric system. A coolant system shall be provided that will supply adequate deionized water to the cutting area. A flow meter shall be provided to ensure proper and uniform flow at the work area. Means shall be provided to recirculate the water through prefilters to remove solid particles. A dielectric cooling system shall be provided to maintain peak performance in temperatures up to 120° Fahrenheit and relative humidity up to 90 percent.

3.4.8 Power supply. A power supply shall be provided with the equipment. It shall be capable of maintaining the optimum feed rate and current setting for varying work thicknesses. Cutting conditions for various workpiece materials can be resident in the power supply control.

3.4.9 Numerical control system. The numerical control system shall be of the software programmable minicomputer CNC type, designed to be selectively operated via on-board programming, manual data input, RS-232 interface, RS-422 interface, or disk drive input. Information stored in memory may be changed by adding, altering or deleting at the discretion of the programmer without alteration or modification to the control. Part program storage shall be in the form of solid state or bubble type memory, Winchester hard disk/flexible disk combination or single/double disk drive units. For those controls having volatile memory, battery power shall be provided to hold memory at full load for a period of not less than 72 hours when the primary source fails. All necessary executive program routines shall be furnished in the form of 1-inch, eight-channel, binary-coded decimal punched tape in conformance with EIA Standard 227-A mini-cassette or floppy disk for controls having volatile memories. When executive programs are supplied in the form of erasable programmable read only memory (EPROM), complete documentation sufficient to interpret and duplicate shall be provided or one additional erasable program read only memory (EPROM) chip shall be provided by the manufacturer. The machine shall be a full 4-axis programmable automatic type control system. The system shall be capable of continuous path, 2-axis (X and Y) or 4-axis (X, Y, U, and V) linear and circular interpolation with provisions for independent UV programming to allow machining of workpiece with dissimilar shapes on top and bottom surfaces. The control shall accept program data and translate this data into positioning and feed motion command signals, as well as initiate the programmed auxiliary and fixed cycles. The control shall provide switchable inch/metric input capability. All axis drives shall accept the translated signals and move the machine slides to the programmed position with the control selecting and monitoring wire feed rate, axis feed rate and path of axis movements as well as the sequence and timing of all operations. All drives shall provide smooth, accurate and immediate response to control commands. The control shall ensure machine functions occur in the proper order and initiate protective action should a faulty condition occur. The control shall utilize solid state, transistorized or integrated circuitry and modular construction. Plug-in circuit boards shall be coded to facilitate identification and location in the system.

3.4.9.1 Data input.

3.4.9.1.1 Punched tape. When specified (see 6.2), the input media shall be 1-inch, 8-channel punched tape. The tape format and codes shall comply with EIA Standards RS-274 and RS-447. The control shall be equipped with a photoelectric tape reader capable of holding, feeding and reading tape prepared in the format specified at a rate not less than 150 characters per second. Unless otherwise specified (see 6.2), the tape reader shall be supplied with standard feed and take up spools 5-1/4 inches in diameter. The tape reader shall be housed in a

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dust free area of the numerical control cabinet and shall be easily accessible for cleaning and tape handling. The control shall be capable of recognizing and acting upon data coded in accordance with either EIA RS-244 or EIA RS-358. The control shall have a tape error detection capability to continuously check for the following errors:

a. Odd parity per EIA RS-244
b. Even parity per EIA RS-358
c. Misaligned sprocket holes
d. No sprocket holes

Any of the above errors shall halt machine functions and illuminate an error light or a message on the data display screen to indicate a fault condition.

3.4.9.1.2 Diskette input. When specified (see 6.2), the input media shall be in the form of 3-1/2 inch diskette. The diskette shall meet the requirements of ANSI/ISO 8860/1, ANSI/ISO 8860/2, and ANSI/ISO 9293. When diskette input is required, the control shall be provided with an integral or portable diskette reader/writer as specified by the procuring activity (see 6.2). The read/writer shall receive and store part program data from the control unit and be able of loading stored programs to the memory of the control unit. The communication between the portable reader/writer shall be by means of EIA RS-232 communication ports.

3.4.9.1.3 Manual data input. A means for manual data input (MDI) shall be furnished to permit manual entry of alphabetical, numerical, and special characters necessary for control of all programmable machine functions. The manually input data shall be immediately displayed on the data display to permit data verification. The MDI shall be capable of entering a block of data into memory upon command.

3.4.9.2 Data display. Unless otherwise specified (see 6.2), a cathode ray tube (CRT) capable of displaying at least 250 characters simultaneously, shall be provided. The CRT shall be capable of displaying as a minimum the following:

a. Four digit block sequence number.
b. Active block of stored data.
c. Actual position of each axis.
d. Feedrate and override condition.
e. Control and system error messages.
f. Graphic depiction of programmed part and cutter path.

3.4.9.3 Modes of operation. The operational modes of the control shall include automatic, single block and MDI.

3.4.9.4 Departure control. The maximum departure permitted by a single command in either incremental or absolute mode shall be not less than plus or minus 99.999 inches or 999.999 mm. The control shall be capable of automatically controlling the acceleration and deceleration of the slides. The resolution of slide movement in either direction shall not be more than 0.0001 inch or 0.001 mm.

3.4.9.5 Absolute/incremental input. The control shall have absolute input capability. This mode shall be selected by a preparatory "G" code which may be switched during the tape program.

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3.4.9.6 Part program edit. A program edit capability shall be provided to allow block insertion, deletion or modification of any part program in the part program memory. Data related to part program edit shall be input by the manual data input provided on the control panel. Edited program data entered into program memory shall be automatically acted upon, in the sequence designated, whenever the part program is cycled.

3.4.9.7 Feed rate override. A feed rate override control shall be provided to allow manual adjustment of the programmed feed rate from zero to not less than 120 percent of the programmed rate in either continuous adjustment or in steps on not more than 10 percent. The feed rate override control shall have a position that allows return of feed rate to that programmed.

3.4.9.8 Sequence number search. The control shall have the capability to permit the operator to search for any desired sequence number.

3.4.9.9 Mirror image The machine shall be capable of producing either right or left hand parts from the same part program through the inversion of X and Y axes. Inversion of each axis shall be by keyboard or switch entry. Mirror image shall be displayed on the data display (CRT).

3.4.9.10 Miscellaneous/preparatory functions. The miscellaneous/preparatory function codes (G and M Codes) shall be in accordance with EIA Standard RS-274.

3.4.9.11 Block delete. The control shall have a block delete feature to permit by-passing of blocks of programmed data.

3.4.9.12 Buffer storage. The control shall have buffer storage for transferring command data from the input media to internal storage without delaying the next incoming command and without interrupting the machine functions.

3.4.9.13 Block-by-block read. The control shall have a block-by-block read function that will allow the operator to cycle the program one block of data at a time.

3.4.9.14 Reversal error compensation. Reversal error compensation shall be provided for each axis of motion to compensate for drive gear backlash when slide reversal occurs. A means shall be provided to input new compensation values when required.

3.4.9.15 Axis calibration. Error compensation shall be provided to correct repeatable errors determined to exist due to leadscrew and feedback device inaccuracies. A means shall be provided to input new compensation values when required.

3.4.9.16 Additional control features. The following additional numerical control features shall be provided:

a. Four quadrant programming.
b. Incremental and continuous jog for each axis
c. Leading zero suppression or decimal point programming
d. Inches per minute (IPM)/Millimeters per minute (MMPM) programming in all modes of operation
e. Edit keylock

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3.4.9.17 Operator control panel. The control unit shall have a control panel which provides as a minimum the following control functions:

a. Emergency stop
b. Cycle start
c. Optional stop
d. Mode selection
e. Feed hold
f. Feedrate override
g. Spindle speed override
h. Jog
i. End-of-block stop

3.4.9.18 Reference zero. The control unit shall have a reference zero to move each axis to a reference to establish synchronization and reset all registers to zero.

3.4.9.19 Set zero. The control unit shall have a "set zero" feature that is software controlled to permit the specified axis register to be reset to zero establishing present axis location as zero.

3.4.9.20 Program proveout. The control shall have a program proveout function that shall allow a program to be run through the control without generating machine motions.

3.4.9.21 Operational software. Operational software shall be resident in the control and shall contain all logic necessary to effectively operate the machine tool. The software shall provide for meaningful diagnostics relating to the utilization of the controls, machine tool, part program, and operator's actions.

3.4.9.22 Programmable interface. A software program for accommodating the softwired interface between the numerical control and the machine shall be provided as part of the executive program or as a separate program in the form of one inch, eight channel punched tape complying with EIA RS-244, or a diskette as appropriate for the input media supplied.

3.4.9.23 Maintenance diagnostics system. Maintenance diagnostics software shall be furnished in either the control as non-volatile memory, or in the form of one inch, eight channel punched tape complying with EIA RS-244, or diskette as appropriate for the input media supplied to provide for diagnostic checking of the machine control unit. The diagnostic software shall test, exercise, and display failures, at least to board level.

3.4.9.24 Peripheral equipment interface. The control shall be equipped with an interface complying with RS-232 and RS-449, or RS-232/RS-449 switchable ports.

3.4.9.25 Incremental feed. The control panel and pendant shall have manual controls for accomplishing both single and continuous, incremental movements of the slides for each controlled axis in both the plus and minus direction. Increment control shall be provided in incremental amounts of 0.0001", 0.001", 0.01", 0.1", and 1.0".

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3.4.10 Electrical requirements. The machine shall be wired to conform to NFPA 79. Unless otherwise specified (see 6.2), each machine shall be designed to operate on 230/460-volts, 3-phase, 60 Hertz (Hz). The machine shall be initially wired for operation as specified (see 6.2).

3.4.11 Motors. Motors shall be rated for continuous duty and shall have ball or roller bearings of the sealed and permanently lubricated type. Unless otherwise specified (see 6.2), alternating-current (AC) motors shall be designed to operate on 60-Hz. The motor enclosure shall meet the requirements for a dripproof enclosure. All motors shall conform to the requirements of NEMA MG-1.

3.4.12 Worklight. When specified (see 6.2), a worklight shall be provided. The light shall have a protective shield, an on-off switch, and shall be adjustable to reflect maximum light on the work area. If a built-in lighting system is provided, the adjustable light described above shall not be required.

3.4.13 Hour meter. When specified (see 6.2), the wire machine shall be equipped with an hour meter. The hour meter shall be installed to display accumulated erosion hours. The meter shall be of the nonresetting type and shall have a range of 0 to 99,999 hours in increments of not greater than 1 hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the machine.

3.5 Size and capacity. Unless otherwise specified (see 6.2), the size and capacity of the wire machine shall meet the requirements of table I for the machine specified. Machine capacity shall be not less than the stated requirements of the size ordered. When a range is given, the required performance shall be from the stated minimum or less to the stated maximum or greater.

3.5.1 Physical size limitations. When specified (see 6.2), the physical size of the machine shall not exceed the height, width, and length restrictions specified by the procuring activity to ensure the machine will fit in its future operating location.

3.6 Performance. The wire machine shall be capable of cutting type D2 tool steel, Rockwell 60 - 62 C scale, 1-inch thick to a tolerance of $\pm$0.0001", total indicator reading (TIR), using a maximum of one rough pass and two finish passes. The workpiece shall have a surface finish of .5 Ra or better. The rough machining rate for this type tool steel shall be no less than 14-inches per hour. Additionally, the wire machine shall be capable of cutting type D2 steel, Rockwell 60-62 C scale, 4-inches thick to a tolerance of $\pm$0.0003 TIR, using a a maximum of one roughing and two finish passes. The workpiece shall have a surface finish of .5 Ra or better. The machining rate for this type of tool steel shall be no less than 3.0 linear/inches per hour. All requirements in table I or not less than stated shall be capable of being met.

3.6.1 Additional performance requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified and fully described (see 6.2).

3.6.2 Taper cut. The machine shall be capable of producing a tapered cut at any angle between 0 and 20^0. The taper shall be automatically oriented along the normal to the cut surface by the CNC system.

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3.7 Standard equipment. All standard equipment normally provided with the manufacturer's commercial machine shall be furnished.

3.8 Optional equipment. Optional equipment shall be furnished as specified and fully described (see 6.2). The optional equipment provided shall be functional without requiring modification of the equipment or machine.

3.9 Marking on instruments, control panels, charts, and plates. All words on instruments, control panels, charts, and plates shall be in the English language. Characters shall be permanently marked in boldface on a contrasting background. All plates shall be corrosion resistant.

3.9.1 Lubrication plate. When specified (see 6.2), a lubrication plate shall be attached to each machine. Unless otherwise specified (see 6.2), the information provided on the plate shall be as listed below.

Points of lubricant application
Servicing interval
Type of lubricant
Federal or military specification for each lubricant (if applicable)

3.9.2 Nameplate. A nameplate shall be securely attached to each machine. Unless otherwise specified (see 6.2), the nameplate shall contain the information listed below. If the wire machine is a special model, the model designation shall include the model of the basic standard wire machine and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Amp rating of largest motor
Short circuit/over-current protection rating
Contract number or order number
National Stock Number or Plant Equipment Code
Date of manufacture

3.10 Workmanship. Workmanship of the wire machine and its accessories shall meet all requirements specified herein.

3.11 Technical data. When technical data (operating manuals, maintenance manuals, parts catalogs, prints, wiring diagrams, lubrication charts, and machine alignment and accuracy test results) is required, it shall be furnished in accordance with the requirements of DD Form 1423. See paragraph 6.3 for additional information.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor

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may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submissions of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

c. Acceptance test (see 4.6).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed in accordance with the test conditions specified in 4.8.1.

4.4 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.7 and all tests in 4.8. The machine shall pass the examination and all tests to be accepted.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each machine prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2), quality conformance inspection shall consist of the examination in 4.7, the tests in 4.8, and the inspection in 4.9. The machine shall pass the examination, all tests, and the inspection to be accepted.

4.6 Acceptance test. Acceptance test shall be performed on each machine to ensure conformance with this specification. Unless otherwise specified (see 6.2), the acceptance test shall be performed only after the wire machine is installed at its final location. The acceptance test shall consist of the examination in 4.7 and all tests in 4.8. The machine shall pass the examination and all tests to be accepted.

4.7 Examination. Each wire machine shall be examined to determine compliance with all requirements of this specification.

4.8 Tests.

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4.8.1 Test conditions. All tests shall be performed in an indoor facility with ambient conditions of 41° to 104° Fahrenheit and 20 to 95 percent relative humidity.

4.8.2 Operational test. Each machine shall be operated at no load for not less than one hour. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period.

4.8.3 Performance tests. The equipment shall be operated in a manner and for a time frame suitable to properly demonstrate and verify the successful operation of the equipment and all of its components parts, relative to the design, construction and performance criteria established herein. All performance tests shall be completed without making any adjustments to or performing any maintenance on the machine.

4.8.3.1 Additional performance tests. Performance tests required in addition to or in lieu of the requirements given in 4.8.3 shall be as specified and fully described (see 6.2).

4.8.4 Maximum metal removal. Using a tool steel workpiece and a wire electrode having a diameter of not less than 0.003", the machine shall demonstrate that it is capable of eroding 0.005 cubic inch of metal per hour per ampere.

4.8.5 Electromagnetic interference control test. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.3.8 using the procedures given in MIL-STD-462.

4.9 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-18058.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The wire EDM covered by this specification is intended for use in shops where electrical discharge machining for uniform tolerances for size and finish on a variety of materials is required. This type machine is versatile in that straight and tapered cuts, external and internal diameters, and rotary faces may be cut in one set-up. The wire machine will be used to perform operations on models, cut complex gears, and reproduce shapes and cavities to a high degree of accuracy.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Size required (see 1.2).

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c. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

d. First article, if required (see 3.1).

e. System of units for measuring and indicating devices (see 3.2.1).

f. Additional safety and health requirements, if required (see 3.2.5).

g. Specify configuration of additional point-of-operation guarding, if required (see 3.2.5).

h. Painting, if different (see 3.3.5).

i. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.8).

j. Specify if punched tape input is required (see 3.4.9.1.1).

k. Specify take up spool size, if different (see 3.4.9.1.1).

l. Specify if diskette input is required (see 3.4.9.1.2).

m. Specify either integral or peripheral diskette reader/writer, if required (see 3.4.9.1.2).

n. Specify data display, if different (see 3.4.9.2).

o. Electrical requirements, if different (see 3.4.10).

p. Specify initial voltage requirements (see 3.4.10).

q. Motors, if different (see 3.4.11).

r. Worklight, if required (see 3.4.12).

s. Hour meter, if required (see 3.4.13).

t. Size and capacity, if different (see 3.5).

u. Specify physical size limitation, if required (see 3.5.1).

v. Additional performance features, if required (see 3.6.1).

w. Specify and fully describe optional equipment, if required (see 3.8).

x. Lubrication plate, if required (see 3.9.1).

y. Information to be provided on lubrication plate, if different (see 3.9.1).

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z. Nameplate, if different (see 3.9.2).

aa. Inspection conditions, if different (see 4.3).

bb. First article inspection, if different (see 4.4).

cc. Quality conformance inspection, if different (see 4.5).

dd. Acceptance test, if different (see 4.6).

ee. Additional performance tests, if required (see 4.8.3.1).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
3.12	DI-TMSS-80527	Commercial Off-The-Shelf (COTS) Manuals.	Require all data be in the English language.
3.12	DI-TMSS-80528	Supplemental data for Commercial Off-The-Shelf Manuals.	Require all data be in the English language.

The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a first article sample, a first production item, or a standard production item from the contractor's current inventory and the number of items to be tested as specified in 4.4. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.5 Definitions.

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6.5.1 Inch-pound units. Inch-pound units are a system of measures based on the yard and pound commonly used in the United States of America and defined by the National Institute of Standards and Technology. Inch-pound units having the same names in other countries may differ in magnitude.

6.5.2 Metric units. Metric units are a system of basic measures defined by the International System of Units based on "Le Systeme International D'Unites (SI)," of the International Bureau of Weights and Measures. These units are described in ASTM E 380 and IEEE 268 (MIL-STD-961).

6.6 Measurement system. In this specification, all measurements, dimensions, sizes, and capacities are given in inch-pound units. These measurements may be converted to metric units through the use of the conversion factors and methods specified in FED-STD-376.

6.7 Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.8 Cross-reference of classification changes. Classification changes from the previous issue of this specification are as follows:

MIL-E-80270	MIL-E-80270A
Size 1	Delete
Size 2	Delete
Size 3	Delete
Size 4	Add
Size 5	Add
Size 6	Add
Size 7	Add
Size 8	Add
Size 9	Add
Size 10	Add
Size 11	Add
Size 12	Add
Size 13	Add

6.9 Safety and health requirements. OSHA 29 CFR 1910 limits only the total hazard level (noise, radiation, electromagnetic emissions, noxious vapors, air contaminants, and heat) of the environment in which a machine will operate. It does not limit the hazard level of individual machines in an operating environment. The procuring activity should analyze the existing hazard level in the proposed operating environment and specify additional requirements necessary to integrate this new machine into its future environment.

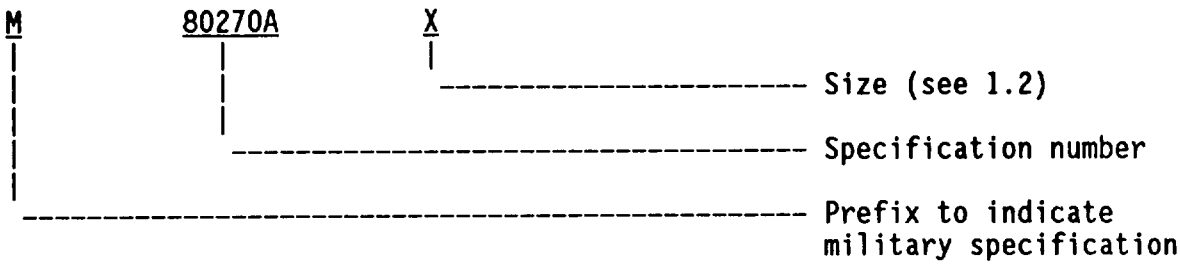
6.10 Training. Training required by the procuring activity should be provided as specified in the contract.

6.11 Warranty. Warranty requirements should be as specified by the procuring activity in the contract.

6.12 Inspection location. The contractor should identify in his response to the solicitation the location where inspection and tests are to be performed.

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6.13 Part or Identifying Number (PIN). The PIN to be used for machines acquired to this specification are created as follows:



(NOTE: See MIL-STD-961C para 4.4a for format.)

6.14 Subject term (keyword) listing.
Dielectric system, coolant system
Taper cut, 0 to 20°
Wire guides, maximum taper 20°
Work table, base

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TABLE I. Sizes and capacities.

Characteristics	Requirements				
	Size 4	Size 5	Size 6	Size 7	Size 8
Table:					
Table dimensions (W x D)	18 x 18	18 x 24	7.5 x 11.5	16.5 x 24.8	23.8 x 25.6
Worktable load capacity	440 lbs.	660 lbs.	200 lbs.	772 lbs.	661.4 lbs.
Table travel	11.8 x 9.8	11.8 x 15	6.5 x 10.5	14.2 x 9.8	11.8 x 13.8
Table speed (in./min)	1.9	1.9	5.3	1.9	1.77
Workpiece dimensions (in.)	16 x 16 x 4	16 x 22 x 4	18 x 27 x 6	20 x 16 x 8	23 x 19 x 6
Wire diameter (in.)	.004 - .012	.004 - .012	.004 - .012	.004 - .012	.004 - .012
Coolant reservoir capacity	52 gal.	52 gal.	40 gal.	52 gal.	60 gal.

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TABLE I. Sizes and capacities continued.

Characteristics	Requirements				
	Size 9	Size 10	Size 11	Size 12	Size 13
Table:					
Table dimensions (W x D)	27.6 x 40.9	23.6 x 31.5	35.6 x 25.5	36.8 x 30.6	55.5 x 50.6
Worktable load capacity	2200 lbs.	1543lbs.	400 lbs.	700 lbs.	2500 lbs.
Table travel	19.7 x 27.6	29.5 x 15.6	11 x 7.5	14 x 12	27 x 15
Table speed (in./min)	1.9	1.77	16	16	16
Workpiece dimensions (in.)	23 x 35 x12	25 x 29 x 8	31 x 22 x 9	34 x 24 x 9	51 x 47 x 9
Wire diameter (in.)	.004 - .012	.004 - .012	.002 - .012	.004 - .012	.004 x .012
Coolant reservoir capacity	52 gal.	80 gal.	55 gal.	65 gal.	100 gal.

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Custodians:
Army - AL
Navy - SH
Air Force - 99

Review activities:
Army - AR, AV
Navy - SA
Air Force - 84

User activities:
Army - ME
Navy - MC

Preparing activity:
DLA - IP

(Project 3410-0050)

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INSTRUCTIONS

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-E-80270A	2. DOCUMENT DATE (YYMMDD) 90-05-02

3. DOCUMENT TITLE
Electrical Discharge Machine, CNC, Wire

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Industrial Plant Equipment Center (DIPEC-SSM)	b. TELEPHONE (Include Area Code) (1) Commercial 775-6229 (2) AUTOVON 683-6229
c. ADDRESS (Include Zip Code) 2163 Airways Boulevard Memphis, TN 38116-5051	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

DD Form 1426, OCT 89 *Previous editions are obsolete.* 198-290

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